Ravi Mangal

Contact Information

CSB 372, CSU
Fort Collins, CO 80523 USA

ravi.mangal@colostate.edu
https://ravimangal.github.io/
Google Scholar

Research Interests

Trustworthy AI, Formal Methods, Assured Autonomy, Program Verification

Education

Georgia Institute of Technology, Atlanta, Georgia, USA
Ph.D., Computer Science — **Dec 2020**
Advisor: Dr. Alessandro Orso

Georgia Institute of Technology, Atlanta, Georgia, USA
M.S., Computer Science — **May 2012**

Veermata Jijabai Technological Institute, Mumbai, India
B.Tech., Information Technology — **May 2010**

Academic Positions

Colorado State University, Fort Collins, Colorado, USA
Assistant Professor, Department of Computer Science — **Aug 2024 – Present**

CyLab, Carnegie Mellon University, Silicon Valley, California, USA
Postdoctoral Researcher with Dr. Corina Păsăreanu — **Jan 2021 – Jun 2024**

Georgia Institute of Technology, Atlanta, Georgia, USA
Graduate Research Assistant — **Jan 2012 – Dec 2020**

Indian Institute of Technology-Bombay, Mumbai, India
Undergraduate Researcher — **May 2008 – Jul 2008**

Honors and Awards

Top Area Chair, NeurIPS Position Paper Track	**2026**
Best paper award nominee at EMSOFT	**2026**
Invited to attend NSF LEVEL UP AI Workshop in Phoenix	**2026**
Distinguished paper award nominee at CAIN	**2025**
Invited to NII Shonan Meeting on Trusted Automatic Programming	**2025**
Amongst the top reviewers at NeurIPS	**2022, 2023, 2024, 2025**
Invited to Dagstuhl Seminar on Resilience and Antifragility of Autonomous Systems	**2024**
Invited to attend the DARPA AI Forward workshop	**2023**
Invited to Dagstuhl Seminar on Machine Learning and Logical Reasoning: The New Frontier	**2022**
Distinguished paper award at ESEC/FSE	**2015**
Distinguished paper award at PLDI	**2014**
Best paper award nominee at ESOP	**2014**

Grants

Semantic Aware Code Generation with LLMs
NSF — *Role: Co-PI*
PI: Limin Jia; Co-PIs: Ravi Mangal, Corina Păsăreanu, Beidi Chen
Total funds awarded: $900,000 (CSU share: $225,000) — **2025 – 2028**

Adversarial Perturbations and Self-Defenses for LLMs on Coding Tasks
IARPA — *Role: Co-PI*
PI: Corina Păsăreanu; Co-PIs: Beidi Chen, Matt Fredrikson, Limin Jia, Ravi Mangal
Total funds awarded: $3,726,055 (CSU share: $405,000) — **2026 – 2027**

LLM Self-Defense Against Adversarial Attacks for Coding Tasks
CyLab Future Enterprise Security Initiative — *Role: PI*
PIs: Corina Păsăreanu, Limin Jia, Ravi Mangal
Total funds awarded: $75,000 — **2023**

Verifiable Personalization for Federated Learning
CyLab Future Enterprise Security Initiative — *Role: PI*
PIs: Corina Păsăreanu, Ravi Mangal
Total funds awarded: $60,000 — **2022**

Advising and Mentoring Experience

Current Students

Rahul Zalkikar (PhD, Colorado State University)	**2026 – Present**
Mikyla Bowen (MS, Colorado State University) *co-advised with Jesse Wilson*	**2026 – Present**
Brendan Dellos (BS, Colorado State University)	**2026 – Present**
Hasan Najib Mahmud (PhD, Colorado State University)	**2025 – Present**
William Scarbro (PhD, Colorado State University) *co-advised with Sanjay Rajopadhye*	**2025 – Present**
Ronaldo Canizales (PhD, Colorado State University)	**2024 – Present**

Former Students

Ben Blount (BS Honors, Colorado State University)	**2025 – 2026**
Md Abdul Hannan (MS, Colorado State University)	**2025 – 2026**
Harshit Singh (MS, Colorado State University)	**2025 – 2026**
Siqi Xie (MS, Carnegie Mellon University)	**2022 – 2023**
Chi Zhang (PhD, Carnegie Mellon University)	**2021 – 2025**
Kartik Sarangmath (BS/MS, Georgia Institute of Technology)	**2019 – 2022**

Research Interns

Ezzedin Amari (Tunisia Polytechnic School – EPT)	**2026**
Davis Davalosh-Delosh (University of Colorado, Boulder)	**2025**

Mentoring Programs

SIGPLAN-M	**2020 – Present**

Research Articles

* indicates equal contribution; (α) indicates alphabetical ordering.

Preprints

[P4] Hasan Najib Mahmud*, Shreya Gupta*, Isha Chaudhary, Nathaniel Enis, **Ravi Mangal**, Gagandeep Singh, and Corina Păsăreanu. A Jagged Frontier: Evaluating Robustness of Code Agents to Semantics-Preserving Transformations. *arXiv preprint arXiv:2608.18389*, 2026

[P3] Ronaldo Canizales, Divya Gopinath, Corina Păsăreanu, and **Ravi Mangal**. Concept-Based Abductive and Contrastive Explanations for Behaviors of Vision Models. *arXiv preprint arXiv:2605.06640*, 2026

[P2] Weichen Yu, **Ravi Mangal**, Terry Yue Zhuo, Matt Fredrikson, and Corina Păsăreanu. A Mixture of Linear Corrections Generates Secure Code. *arXiv preprint arXiv:2507.09508*, 2025

[P1] **Ravi Mangal***, Klas Leino*, Zifan Wang*, Kai Hu*, Weichen Yu, Corina Păsăreanu, Anupam Datta, and Matt Fredrikson. Is Certifying ℓ_p Robustness Still Worthwhile? *arXiv preprint arXiv:2310.09361*, 2023

Journal Articles

[J2] (α) Radu Calinescu, Calum Imrie, **Ravi Mangal**, Genaína Nunes Rodrigues, Corina Păsăreanu, Misael Alpizar Santana, and Gricel Vázquez. Controller Synthesis for Autonomous Systems

with Deep-Learning Perception Components. *IEEE Transactions on Software Engineering (TSE)*, 2024

[J1] Klas Leino*, Chi Zhang*, **Ravi Mangal***, Matt Fredrikson, Bryan Parno, and Corina Păsăreanu. Degradation Attacks on Certifiably Robust Neural Networks. *Transactions on Machine Learning Research (TMLR)*, 2022

Conference Papers

[C25] William Scarbro and **Ravi Mangal**. Interval POMDP Shielding for Imperfect-Perception Agents. In *International Conference on Embedded Software (EMSOFT)*, 2026
★ **Best Paper Award Nominee**

[C24] Weichen Yu, **Ravi Mangal**, Yinyi Luo, Kai Hu, Jingxuan He, Corina Păsăreanu, and Matt Fredrikson. SecCodePRM: A Process Reward Model for Code Security. In *International Conference on Machine Learning (ICML)*, 2026

[C23] Yibo Peng*, James Song*, Lei Li*, Xinyu Yang, Mihai Christodorescu, **Ravi Mangal**, Corina Păsăreanu, Haizhong Zheng, and Beidi Chen. When "Correct" Is Not Safe: Can We Trust Functionally Correct Patches Generated by Code Agents? In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026

[C22] William Scarbro*, Calum Imrie*, Sinem Getir Yaman, Kavan Fatehi, Corina Păsăreanu, Radu Calinescu, and **Ravi Mangal**. Conformal Safety Shielding for Imperfect-Perception Agents. In *International Conference on Runtime Verification (RV)*, 2025

[C21] Christopher Watson, Rajeev Alur, Divya Gopinath, **Ravi Mangal**, and Corina Păsăreanu. Scenario-Based Compositional Verification of Autonomous Systems with Neural Perception. In *International Symposium on AI Verification (SAIV)*, 2025

[C20] Nils Palumbo, **Ravi Mangal**, Zifan Wang, Saranya Vijayakumar, Corina Păsăreanu, and Somesh Jha. Validating Mechanistic Interpretations: An Axiomatic Approach. In *International Conference on Machine Learning (ICML)*, 2025

[C19] Felipe Toledo, Sebastian Elbaum, Divya Gopinath, Ramneet Kaur, **Ravi Mangal**, Corina Păsăreanu, Anirban Roy, and Susmit Jha. Monitoring Safety Properties for Autonomous Driving Systems with Vision-Language Models. In *Engineering Reliable Autonomous Systems (ERAS)*, 2025

[C18] Qiulu Peng, Chi Zhang, **Ravi Mangal**, Corina Păsăreanu, and Limin Jia. Random Perturbation Attack on LLMs for Code Generation. In *International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2025

[C17] Boyue Caroline Hu, Divya Gopinath, Corina Păsăreanu, Nina Narodytska, **Ravi Mangal**, and Susmit Jha. Debugging and Runtime Analysis of Neural Networks with VLMs (A Case Study). In *International Conference on AI Engineering – Software Engineering for AI (CAIN)*, 2025
★ **Distinguished Paper Award Nominee**

[C16] Chi Zhang, Zifan Wang, Ruoshi Zhao, **Ravi Mangal**, Matt Fredrikson, Limin Jia, and Corina Păsăreanu. Attacks and Defenses for Large Language Models on Coding Tasks. In *IEEE/ACM International Conference on Automated Software Engineering: New Ideas and Emerging Results (ASE-NIER)*, 2024

[C15] **Ravi Mangal**, Nina Narodytska, Divya Gopinath, Boyue Caroline Hu, Anirban Roy, Susmit Jha, and Corina Păsăreanu. Concept-Based Analysis of Neural Networks via Vision-Language Models. In *International Symposium on AI Verification (SAIV)*, 2024

[C14] Corina Păsăreanu, **Ravi Mangal**, Divya Gopinath, and Huafeng Yu. Assumption Generation for the Verification of Learning-Enabled Autonomous Systems. In *International Conference on Runtime Verification (RV)*, 2023

[C13] Corina Păsăreanu, **Ravi Mangal**, Divya Gopinath, Sinem Getir Yaman, Calum Imrie, Radu

Calinescu, and Huafeng Yu. Closed-Loop Analysis of Vision-Based Autonomous Systems: A Case Study. In *International Conference on Computer Aided Verification (CAV)*, 2023

[C12] **Ravi Mangal***, Zifan Wang*, Chi Zhang*, Klas Leino, Corina Păsăreanu, and Matt Fredrikson. On the Perils of Cascading Robust Classifiers. In *International Conference on Learning Representations (ICLR)*, 2023

[C11] (α) Divya Gopinath, Luca Lungeanu, **Ravi Mangal**, Corina Păsăreanu, Siqi Xie, and Huafeng Yu. Feature-Guided Analysis of Neural Networks. In *International Conference on Fundamental Approaches to Software Engineering (FASE)*, 2023

[C10] **Ravi Mangal**, Kartik Sarangmath, Aditya V. Nori, and Alessandro Orso. Probabilistic Lipschitz Analysis of Neural Networks. In *International Static Analysis Symposium (SAS)*, 2020

[C9] **Ravi Mangal**, Aditya V. Nori, and Alessandro Orso. Robustness of Neural Networks: A Probabilistic and Practical Approach. In *International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER)*, 2019

[C8] Sulekha Kulkarni, **Ravi Mangal**, Xin Zhang, and Mayur Naik. Accelerating Program Analyses by Cross-Program Training. In *ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*, 2016

[C7] **Ravi Mangal**, Xin Zhang, Aditya Kamath, Aditya V. Nori, and Mayur Naik. Scaling Relational Inference Using Proofs and Refutations. In *AAAI Conference on Artificial Intelligence (AAAI)*, 2016

[C6] Xin Zhang, **Ravi Mangal**, Aditya V. Nori, and Mayur Naik. Query-Guided Maximum Satisfiability. In *ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, 2016

[C5] **Ravi Mangal**, Xin Zhang, Aditya V. Nori, and Mayur Naik. Volt: A Lazy Grounding Framework for Solving Very Large MaxSAT Instances. In *International Conference on Theory and Applications of Satisfiability Testing (SAT)*, 2015

[C4] **Ravi Mangal**, Xin Zhang, Aditya V. Nori, and Mayur Naik. A User-Guided Approach to Program Analysis. In *ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2015
★ **ACM SIGSOFT Distinguished Paper Award**

[C3] Xin Zhang, **Ravi Mangal**, Mayur Naik, and Hongseok Yang. Hybrid Top-Down and Bottom-Up Interprocedural Analysis. In *ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2014

[C2] Xin Zhang, **Ravi Mangal**, Radu Grigore, Mayur Naik, and Hongseok Yang. On Abstraction Refinement for Program Analyses in Datalog. In *ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2014
★ **Distinguished Paper Award**

[C1] **Ravi Mangal**, Mayur Naik, and Hongseok Yang. A Correspondence Between Two Approaches to Interprocedural Analysis in the Presence of Join. In *European Symposium on Programming (ESOP)*, 2014
★ **Best Paper Award Nominee**

Invited Papers

[I1] Sayan Mitra, Corina Păsăreanu, Pavithra Prabhakar, Sanjit A. Seshia, **Ravi Mangal**, Yangge Li, Christopher Watson, Divya Gopinath, and Huafeng Yu. Formal Verification Techniques for Vision-Based Autonomous Systems – A Survey. In *Principles of Verification: Cycling the Probabilistic Landscape. Essays Dedicated to Joost-Pieter Katoen on the Occasion of His 60th Birthday*. Springer, 2024

Workshop Papers and Abstracts

[W5] Md Abdul Hannan*, Ronghao Ni*, Chi Zhang, Limin Jia, **Ravi Mangal**, and Corina Păsăreanu. On the Difficulty of Selecting Few-Shot Examples for Effective LLM-Based Vulnerability Detection. In *LLM Assisted Security and Trust Exploration Workshop (LAST-X)*, 2026

[W4] **Ravi Mangal** and Corina Păsăreanu. A Cascade of Checkers for Run-Time Certification of Local Robustness. In *Workshop on Formal Methods for ML-Enabled Autonomous Systems (FoMLAS)*, 2022

[W3] Klas Leino, Aymeric Fromherz, **Ravi Mangal**, Matt Fredrikson, Bryan Parno, and Corina Păsăreanu. Self-Correcting Neural Networks for Safe Classification. In *Workshop on Formal Methods for ML-Enabled Autonomous Systems (FoMLAS)*, 2022

[W2] **Ravi Mangal**, Aditya V. Nori, and Alessandro Orso. Checking Probabilistic Properties of Neural Networks via Symbolic Methods and Sampling. In *ICSE Workshop on Testing for Deep Learning and Deep Learning for Testing (DeepTest)*, 2019

[W1] **Ravi Mangal**, David Devecsery, and Alessandro Orso. On Optimally Combining Static and Dynamic Analyses for Intensional Program Properties. In *Southeast Regional Programming Languages Seminar (SERPL)*, 2019

Technical Reports

[T1] **Ravi Mangal**, Xin Zhang, Mayur Naik, and Aditya V. Nori. Solving Weighted Constraints with Applications to Program Analysis. Technical Report GT-CS-15-03, Georgia Institute of Technology, 2015

Research Talks

Invited Talks

Explaining AI Model Behavior Through the Concepts in Its Internal Representations
AI Day with Microway and NVIDIA, Data Science Research Institute, CSU **Oct 2026**

AI Coding Assistants: Security Risks and Defenses
CONIA **Oct 2025**

Concept-based Semantic Analysis of DNNs
University of Wyoming **Apr 2025**

Analyzing Safety of Learning-enabled Systems: Bottom-up and Top-down Perspectives
University of Colorado, Boulder **Nov 2024**

Closed-loop Analysis of Vision-based Autonomous Systems: A Case Study
Cohere for AI **May 2024**

Safety Analysis of Vision-based Autonomous Systems
CMU CyLab Partners Conference **Oct 2023**

Feature-Guided Engineering of Neural Networks
CMU CyLab Partners Conference **Oct 2022**

The Necessity of Run-time Techniques for Safe ML
Dagstuhl Seminar on Machine Learning and Logical Reasoning: The New Frontier **Jul 2022**

Repairing Neural Classifiers at Run-time: Safety for Free
CMU CyLab Partners Conference **Oct 2021**

A User-Guided Approach to Program Analysis
Microsoft Research India **Aug 2015**

Conference and Workshop Presentations and Posters
SAIV'24, CAV'23, FoMLAS'22, SAS'20, ICSE-NIER'19, DeepTest'19, SERPL'19, AAAI'16, SAT'15, FSE'15, ESOP'14

Teaching Experience

Instructor

CS580B1 Trustworthy AI	**Fall 2026**
CS445 Introduction to Machine Learning	**Spring 2026**
CS454 Principles of Programming Languages	**Spring 2025**
CS580B1 Trustworthy Machine Learning	**Fall 2024**

Teaching Assistant

CS6340 Software Analysis and Testing (Online)	**Spring 2016**
CS6340 Software Analysis and Testing	**Spring 2016**
CS8803 Foundations of Programming Languages	**Fall 2013**
CS8803 Foundations of Programming Languages	**Fall 2012**
CS6340 Software Analysis and Testing	**Fall 2011**

PhD and Masters Thesis Committees

PhD

Ryan Couture	Colorado State University
Roya Daneshi	Colorado State University
Rupasree Dey	Colorado State University
Eric Enos	Colorado State University
Md Fazle	Colorado State University
Bassem Ghorbel	Colorado State University
Carine Graff	Colorado State University
Md Nazmul Islam	Colorado State University
Ibrahim Khebour	Colorado State University
Jim Muth	Colorado State University
Soumyadip Roy	Colorado State University
Lucas V. Vieira	Colorado State University
Austin Youngren	Colorado State University
Chi Zhang	Carnegie Mellon University

Masters

Nhat Chu	Colorado State University
Lucas Dowers	Colorado State University
Eric Redman	Colorado State University
Siqi Xie	Carnegie Mellon University
Jonathon Yallop	Colorado State University

Academic Service

Organizing Committees

Publicity Chair, SAIV'27	**2027**
Local Arrangements Co-Chair, PLDI'26	**2026**

Area Chair
NeurIPS'26 (Position Paper Track; **Top Area Chair**)

Program Committees
2027: ICSE SEIP, AAAI
2026: ASE, PLDI SRC, SAIV, NFM, CAIN, RAIE, OOPSLA, FASE
2025: CAV, SAIV, ASQAP, RAIE, VMCAI, AAAI, ICSE
2024: FM, FMCAD, SAIV, IJCAI, NFM, CAIN, CAV

Reviewer

Journals: IEEE TPAMI, TOPLAS, IEEE Software, IEEE Security & Privacy, ACM Journal on Responsible Computing, International Journal of Information Security, TOSEM, Journal of AI Research, IEEE Transactions on Big Data, TMLR
Conferences: NeurIPS'22–'25 (**Top Reviewer** each year), ICML'24, ICML'25, ICLR'22, ICLR'24
Funding Agencies: NSF Panels (2022, 2025, 2026), NWO

Sub-Reviewer
NFM'23, POPL'22, CAIN'22, Oakland'22, PLDI'21, ICSE'20, ISSTA'20, FSE'19, FSE'18, ISSTA'18, SPIN'17, RV'17, JCST'17, ESSOS'17, CAV'14, HVC'14, ICSE-SRC'14

Artifact Evaluation Committees
OOPSLA'24, POPL'23, POPL'20, ISSTA'18, OOPSLA'17, OOPSLA'16

Department Service (CS, Colorado State University)
Graduate Recruiting Committee **Aug 2025 – Present**
Operations Committee **Aug 2024 – Jul 2025**

Industry Experience

Microsoft Research, Redmond, Washington, USA
Research Intern **May 2016 – Aug 2016**

Google, Mountain View, California, USA
Research Intern **May 2014 – Aug 2014**

Nvidia, Santa Clara, California, USA
Software Intern **May 2011 – Aug 2011**

Microsoft, Hyderabad, India
Software Development Engineer in Test Intern **May 2009 – Jul 2009**

Last updated: October 2026